

Food Access & the CTA Red Line Extension:

An Economic Analysis of Food Deserts in Roseland, Chicago

ECO 575 Practicum Team · DePaul University*

Executive Summary · Spring 2026

OVERVIEW

Roseland, a historic community on Chicago's Far South Side, faces one of the city's most severe food access crises. This report summarizes a semester-long study by DePaul University's MS in Economics and Quantitative Analysis students examining why Roseland lacks adequate grocery access, what forces drive that gap, and how the CTA's Red Line Extension, currently under construction, can serve as a catalyst for change. The findings and recommendations are addressed to the Chicago Transit Authority and the Roseland Chamber of Commerce.

The data paint a stark picture. Roseland's food desert rate, the share of census tracts where residents are simultaneously low-income and more than half a mile from the nearest supermarket, is 3.8 times the citywide average. Median household income runs 37 percent below Chicago's citywide figure, and SNAP participation is 75 percent higher than the city average. These are not marginal differences; they reflect a community that has been systematically underserved by the private food retail market for decades.

WHAT THE DATA SHOWS: THE ROOT CAUSES OF ROSELAND'S FOOD GAP

To understand why the food gap exists, the team tested four widely cited theories about what drives grocery store location decisions. The results were sometimes counterintuitive.

Crime Is Not the Driver

Despite Roseland's elevated crime rate, the data show that crime is not a significant predictor of grocery store presence across Chicago neighborhoods. Grocery stores operate in many high-crime communities throughout the city. Crime and food deserts tend to appear together because both are products of the same underlying conditions, concentrated poverty and decades of disinvestment, not because one causes the other.

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Income and Population Density Are the Real Predictors

The two factors with the strongest relationship to grocery store presence are household income and population density. Median income is statistically linked to the number of grocery stores per square mile, and Roseland's income gap relative to the rest of Chicago is large enough to make private retail investment unviable without public support. On density, the data show that the most densely populated neighborhoods in Chicago (in terms of people per square mile) have 2.9 more grocery stores per square mile than the least dense and are 10 percent less likely to be classified as food deserts. Roseland, with approximately 10,000 residents per square mile, falls well below the trend line, meaning it has even fewer stores than neighborhoods of comparable density elsewhere in the city. Some of this may reflect that Roseland is surrounded by other less-densely populated neighborhoods, decreasing the total number of potential customers for any grocery store in the area.

Race Reflects Systemic Disinvestment

As the share of Black residents in a Chicago neighborhood rises, grocery store presence falls, even after controlling for income and density. Roseland is 97 percent Black. Its food gap is inseparable from the long history of redlining, discriminatory commercial lending, and deliberate retail withdrawal from predominantly Black communities on Chicago's South and West Sides. This finding matters for policy: market-based solutions alone, such as tax credits and financing incentives for private grocery chains, are unlikely to be sufficient. The federal Healthy Food Financing Initiative has succeeded at getting stores built nationally but has produced little evidence of sustained dietary improvement. Addressing Roseland's food gap will require both market incentives and structural public investment.

HOW THE RED LINE EXTENSION CHANGES THE EQUATION

The CTA Red Line Extension, a new 5.6-mile corridor with four stations reaching 130th Street, currently under construction, is the most significant public infrastructure investment in Roseland in a generation. It has the potential to directly address the root causes of the food gap through three interconnected effects.

New transit access increases population density. Research on U.S. rail extensions finds 2 to 5 percent population growth within a half-mile of new stations over a decade, contingent on supportive zoning that allows higher-density residential development near station areas. The RLE is projected to carry approximately 40,000 trips per day by 2040, a scale of new activity that signals to grocery retailers a viable and growing customer base in Roseland for the first time in decades.

The extension also creates jobs directly and expands job access for Roseland residents. Construction alone is estimated to generate over 12,500 direct jobs and 59,800 indirect jobs. Beyond construction, the RLE gives Roseland residents faster, more reliable access to employment centers across the city. An estimated 25,000 additional jobs become reachable for community members who previously faced long, unreliable commutes, expanding household incomes and economic stability for the families who need it most.

These gains in density, employment, and income will in turn be reflected in rising property values. Evidence from comparable transit investments shows that property values within a half-mile of new rail stations rise substantially in the years surrounding opening; Chicago's Orange Line saw a 17 percent increase in home values near station locations, generating approximately \$216 million in property gains. Rather than viewing this as a risk, it should be read as confirmation: when more people want to live in Roseland, when more residents are working and earning, and when the neighborhood becomes more connected to the city, the community's economic foundation strengthens in ways that make lasting retail investment, including the grocery stores Roseland has long been denied, a realistic prospect for the first time in a generation.

These gains will not automatically produce food access. Left to market forces, new transit corridors tend to attract dollar stores, fast food, and liquor retail. Intentional planning and public incentives are required to turn transit access into food access. Additionally, the benefits of the RLE are likely to take several years to materialize, with the RLE not scheduled to be completed until 2030. Thus, the recommendations we propose address both shorter-term solutions to address immediate food access challenges and longer-term solutions that coincide with the introduction of the RLE.

FOUR RECOMMENDATIONS FOR THE CTA AND THE ROSELAND CHAMBER

The team developed four phased recommendations spanning from actions achievable in weeks to long-term neighborhood transformation anchored to Red Line station development.

1. Grocery Delivery and Pop-Up Fresh Markets: Immediate

The most urgent need is getting fresh food to residents who cannot wait years for a new store. This means launching a grocery delivery program, modeled on the Dream Deliveries initiative, with a five-dollar flat SNAP/EBT delivery fee and local hires at living wages, alongside pop-up produce markets hosted at community institutions such as Roseland Christian Ministries and local churches, using LINK Match to double the purchasing power of SNAP dollars. The Fresh Moves model, which retrofits CTA buses as mobile markets, provides a direct template. Startup costs are in the low thousands and are fundable through existing ARPA grants and corporate sponsorships.

2. Corner Store Nutrition Incentives and Produce Prescriptions: Near-Term

Roseland does not need a new store to expand fresh produce access immediately. The Warehouse Food Market at 538 W 119th Street, an 18-year community institution already trusted for fresh meat and bulk staples, carries a limited produce selection that federal programs can help close. The team recommends enrolling The Warehouse in Link Up Illinois, which doubles SNAP dollars spent on produce at participating stores, and VeggieRx, through which physicians can prescribe free weekly produce boxes to patients with diet-related conditions. Both programs hold active federal GusNIP grants and have established reporting infrastructures. Roseland does not need to build new nonprofit capacity, it simply needs to be added as a redemption site

to the existing grants. Federal reimbursement could cover the cost.

3. A Grocery-Focused Demand-Responsive Transit Pilot: 2027 to 2029

Transportation is as important as proximity. Seniors, households with young children, and residents without cars face real barriers to reaching a full-service store and returning with a week's worth of food. The team recommends establishing a shared-ride grocery shuttle connecting Roseland residents to Walmart, Jewel-Osco, and Food 4 Less on weekend mornings, modeled on Flint, Michigan's Ride to Groceries program. Pooled vans keep costs near \$25 per ride, below the current Pace ADA rate of \$40, with fares set at \$3.25 and free for SNAP households. Our estimates suggest that the pilot costs between \$130,000 to \$585,000 per year, depending upon the hours of availability, and may be fundable through Illinois Senate Bill 2111 and Federal Section 5310 grants. It is designed from the outset to fold into permanent Red Line Extension service after 2029, feeding riders to the new stations rather than competing with them.

4. Station-Area Grocery Development at Michigan Avenue: Long-Term

The long-term anchor is a full-service community-owned grocery store developed as part of the mixed-use station-area development planned around the Michigan Avenue stop. The 1Fifteen project, a \$48.3 million development on two acres of city-owned land adjacent to the planned station, including a 15,000 square foot anchor grocery store, an 8,000 square foot market hall, and 58 affordable homes, is already underway and provides the immediate blueprint. A parallel community-run co-op, modeled on IMAN's Go Green Community Fresh Market in Englewood, can be developed for approximately \$5 million using the Illinois Grocery Initiative, Healthy Food Financing Initiative grants, and New Markets Tax Credits, the same funding stack that enabled Forty Acres Fresh Market to secure \$9 million in NMTC and \$2.5 million from the City of Chicago. A co-op structure gives residents an ownership stake and ensures profits are reinvested locally.

WHAT WE ARE ASKING FOR

From the CTA: Embed grocery-first retail requirements into station-area planning for the 111th Street and Michigan Avenue stations. Use the 1Fifteen project as the template for remaining city-owned parcels along the corridor and make fresh food access an explicit criterion in requests for proposals issued to private developers.

From the Roseland Chamber: Champion The Warehouse Food Market's enrollment in Link Up Illinois and VeggieRx as the immediate, low-cost first step. Mobilize member businesses and community institutions as venues for pop-up markets and partners in the grocery transit pilot.

From both, working together: Align the financing tools that already exist, Tax Increment Financing, the Small Business Improvement Fund, DCEO retail grants from \$150,000 to \$2.4 million, and INVEST South/West, which has already catalyzed \$1.4 billion in investment on Chicago's South and West Sides, around the RLE station corridor. No new programs are needed. What is needed is coordination and the deliberate commitment to make food access central to how the Red Line Extension reshapes Roseland.

Roseland's food gap is the cumulative result of decades of racial disinvestment, market failure, and public neglect. The Red Line Extension opens a generational window to correct that history. The question is whether the institutions with the power to act will seize it.

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